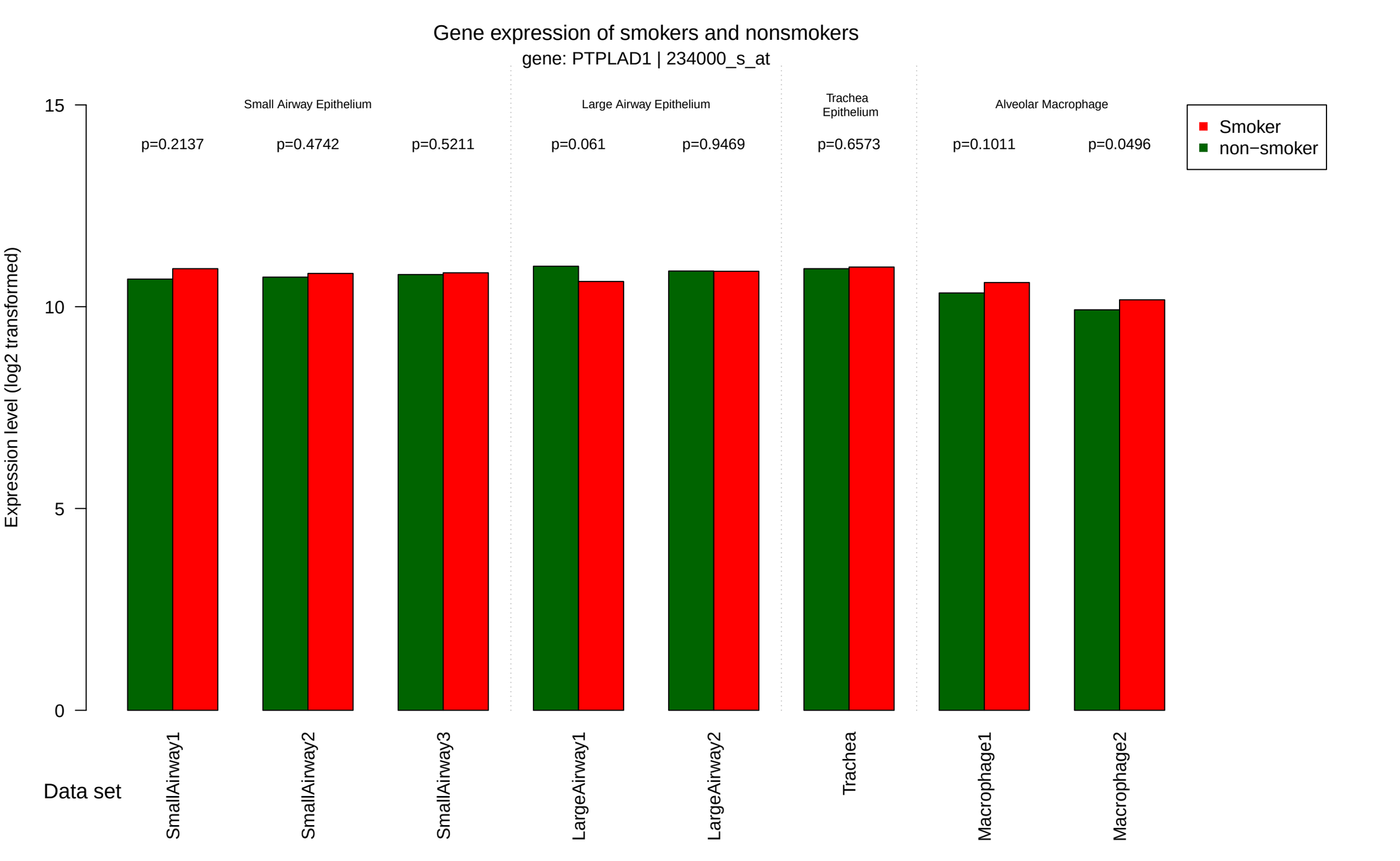
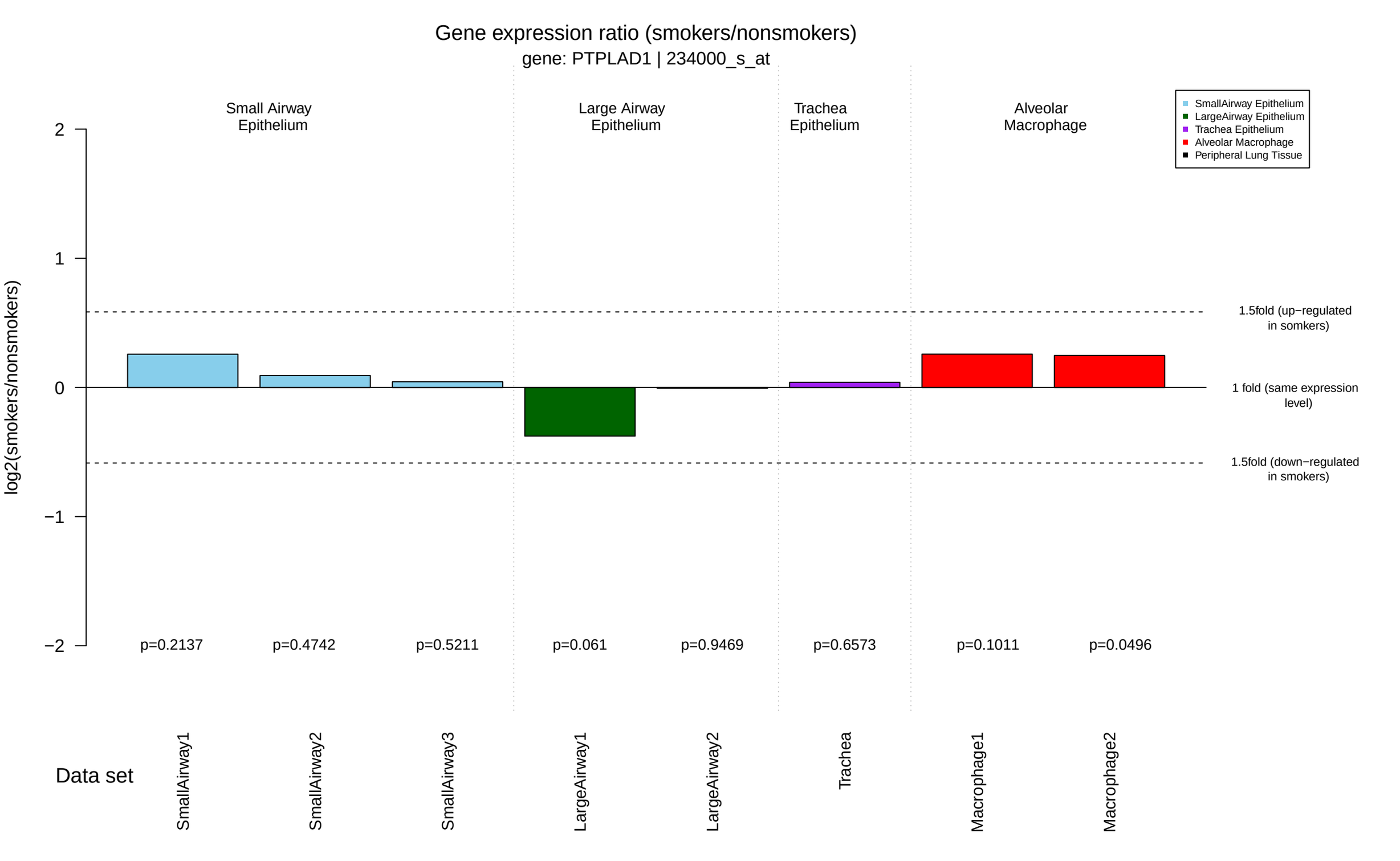
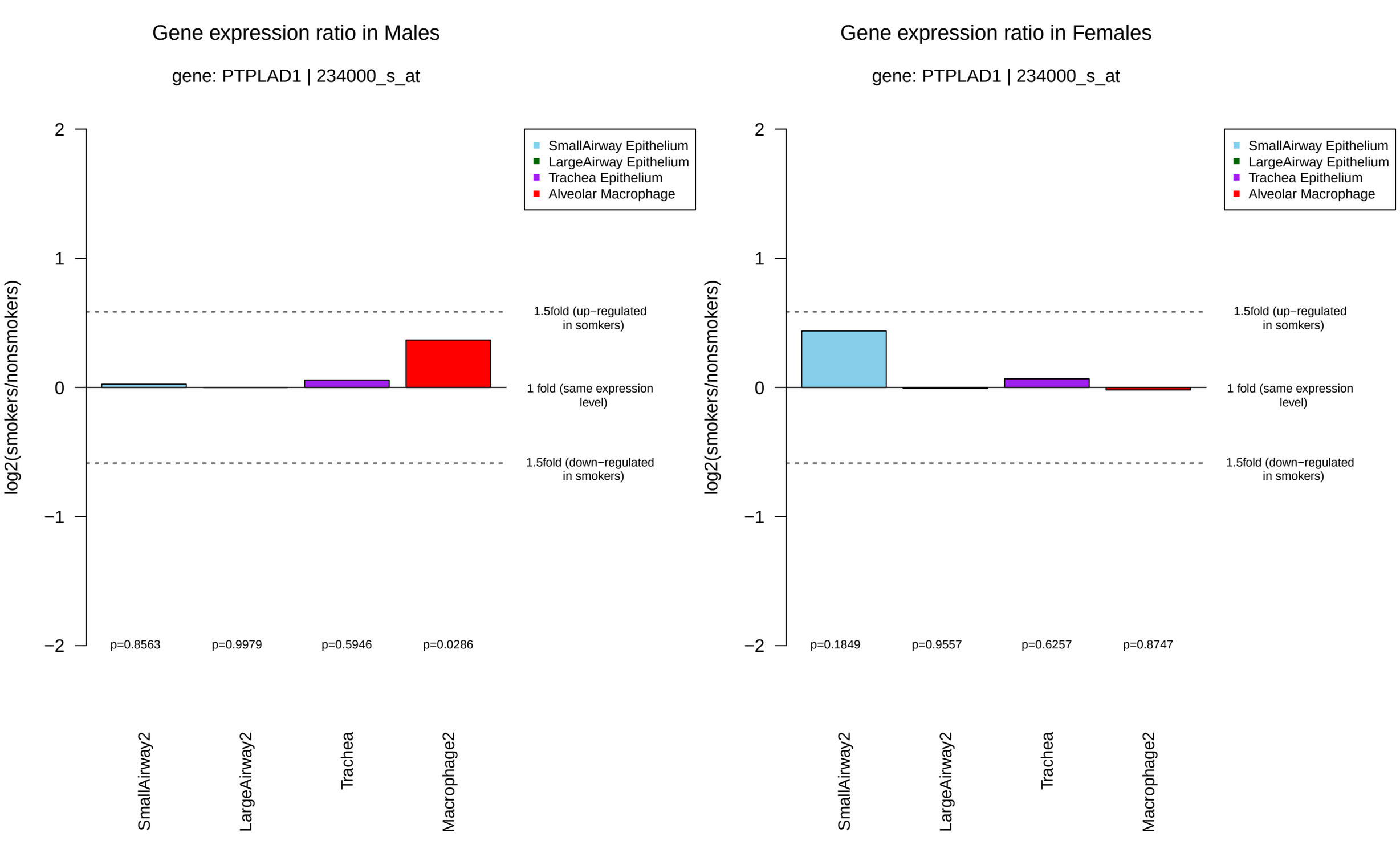
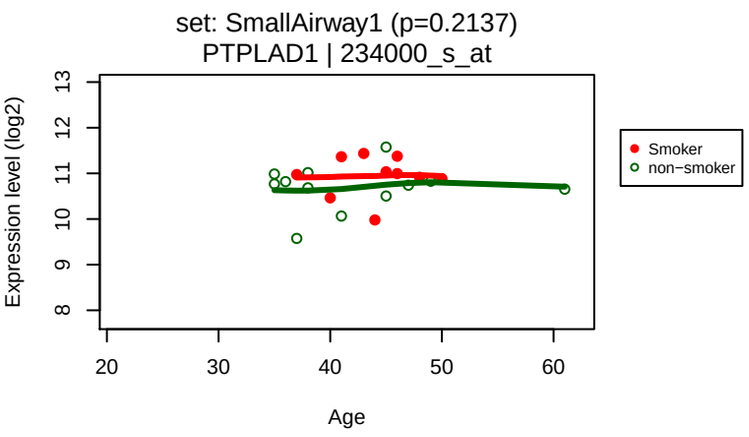
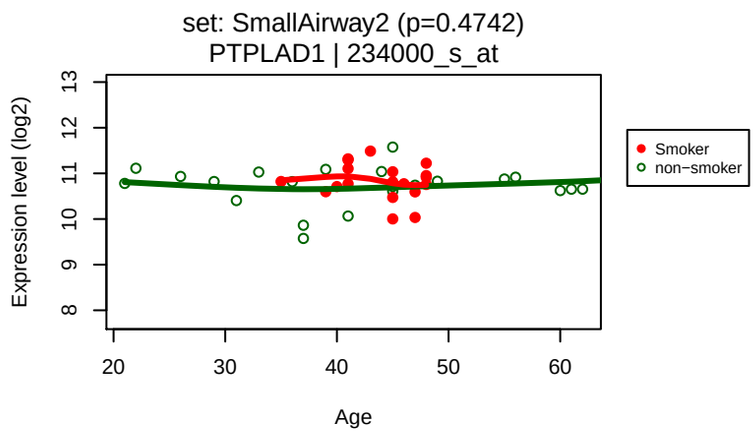
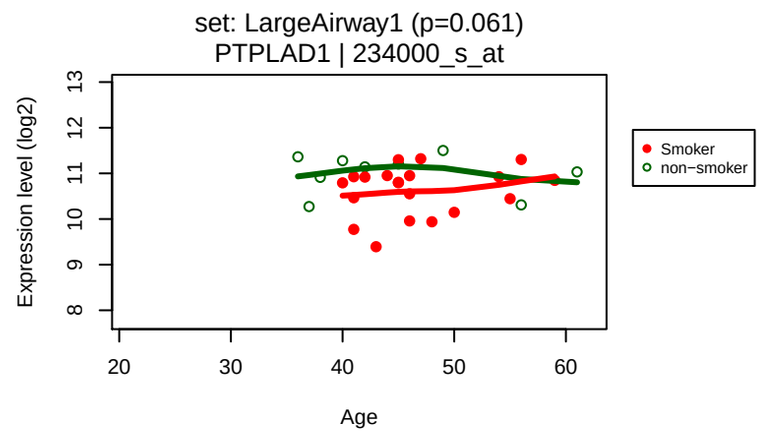

Gene expression of smokers and nonsmokers
gene: PTPLAD1 | 234000_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2137
p=0.4742
p=0.5211
p=0.061
p=0.9469
p=0.6573
p=0.1011
p=0.0496
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

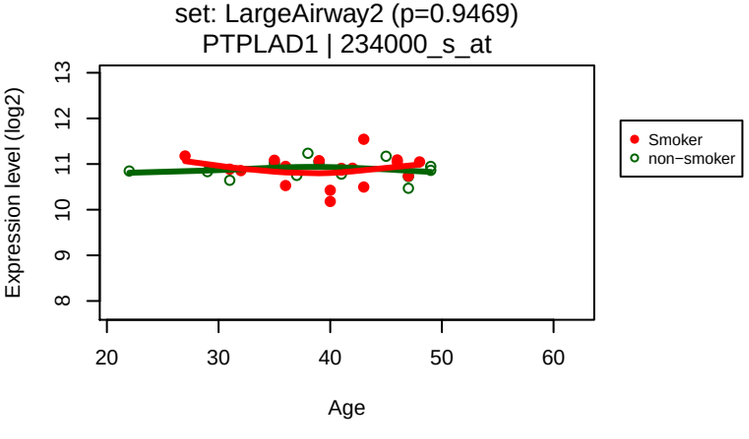

Gene expression ratio (smokers/nonsmokers)
gene: PTPLAD1 | 234000_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2137
p=0.4742
p=0.5211
p=0.061
p=0.9469
p=0.6573
p=0.1011
p=0.0496
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

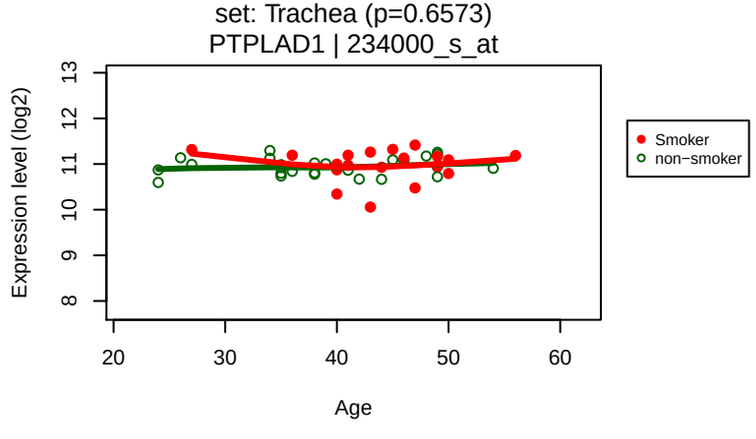

Gene expression ratio in Males
gene: PTPLAD1 | 234000_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8563
p=0.9979
p=0.5946
p=0.0286
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTPLAD1 | 234000_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1849
p=0.9557
p=0.6257
p=0.8747
SmallAirway2
LargeAirway2
Trachea
Macrophage2

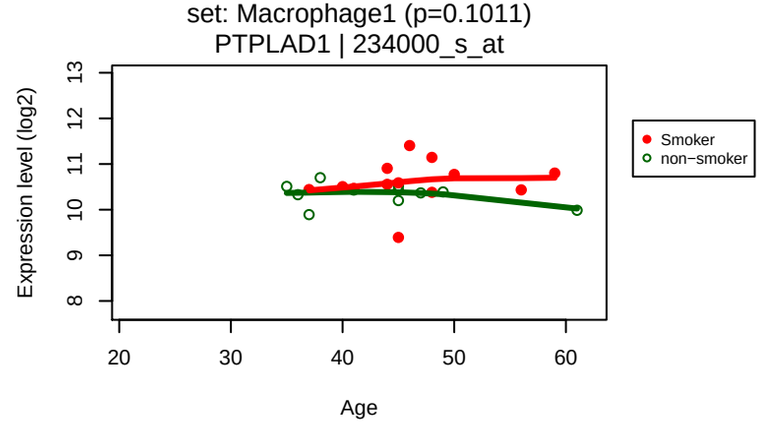

set: SmallAirway1 (p=0.2137)
PTPLAD1 | 234000_s_at
Expression level (log2)
Age
Smoker
non−smoker

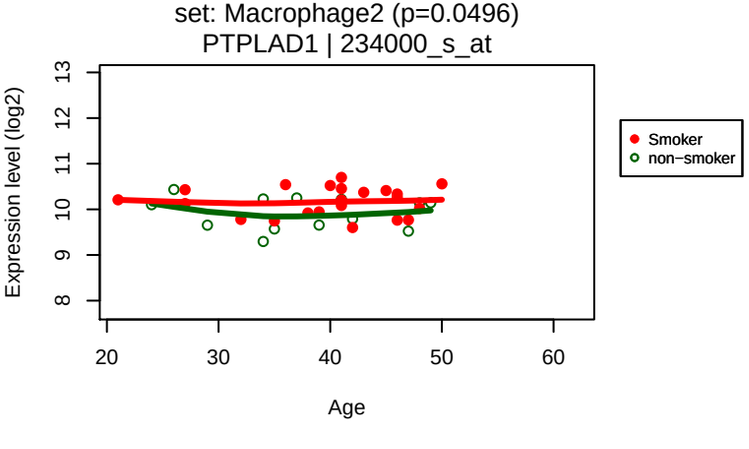

set: SmallAirway2 (p=0.4742)
PTPLAD1 | 234000_s_at
Expression level (log2)
Age
Smoker
non−smoker

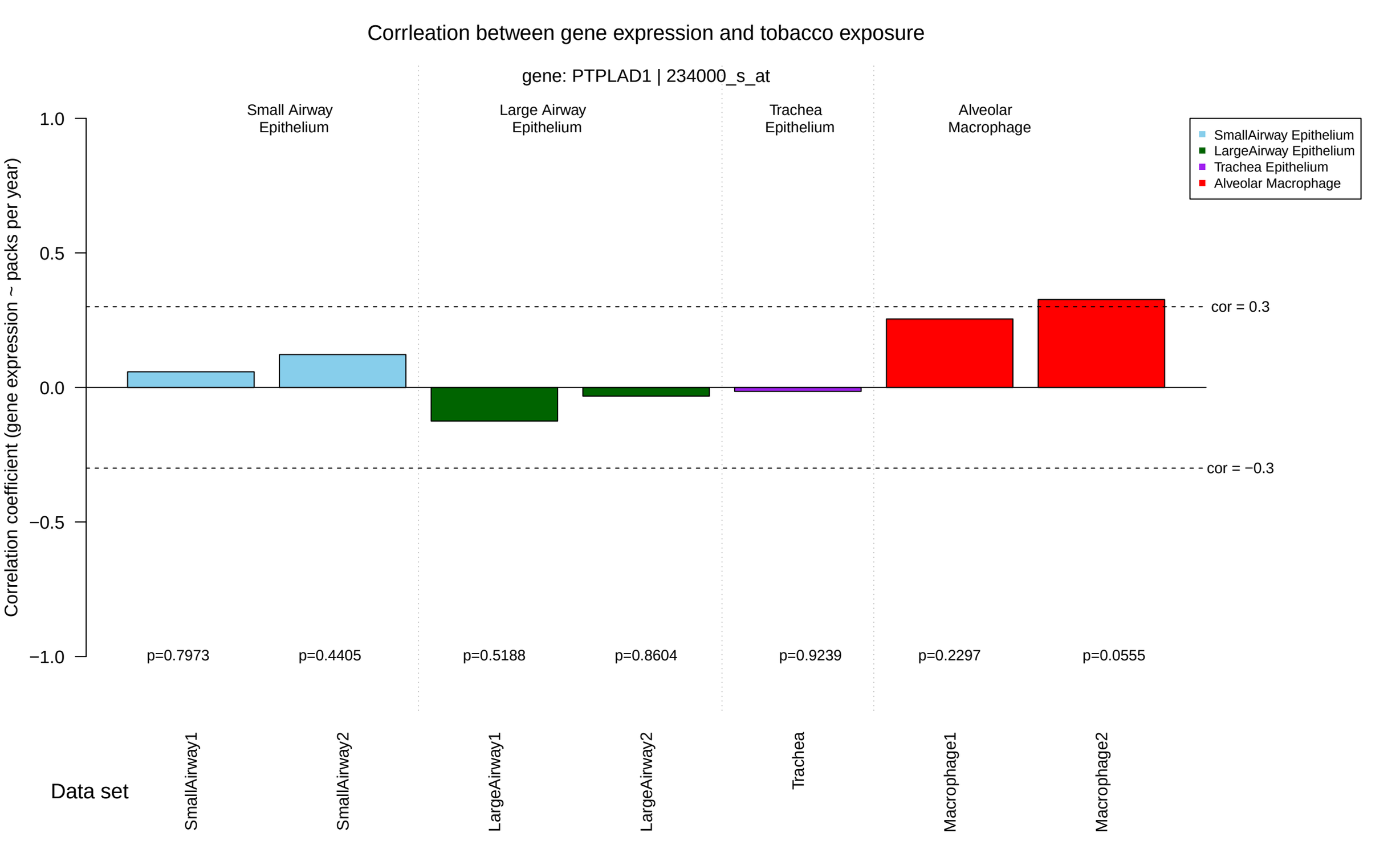

set: LargeAirway1 (p=0.061)
PTPLAD1 | 234000_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9469)
PTPLAD1 | 234000_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6573)
PTPLAD1 | 234000_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1011)
PTPLAD1 | 234000_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0496)
PTPLAD1 | 234000_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTPLAD1 | 234000_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7973
p=0.4405
p=0.5188
p=0.8604
p=0.9239
p=0.2297
p=0.0555
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2